VMC Vidyamandir Classes SINCE 1986

IIT JEE | MEDICAL | FOUNDATION

CODE A

JEE Main - 6 | JEE 2022 (Gen 1 & 2)

Date: 04/01/2021 **Maximum Marks: 300**

Timing: 04:00 PM - 09:00 PM **Duration: 3.0 Hrs**

General Instructions

1. The test is of **3 hours** duration.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them**.
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) : _

Roll Number : _

OMR Bar Code Number : _

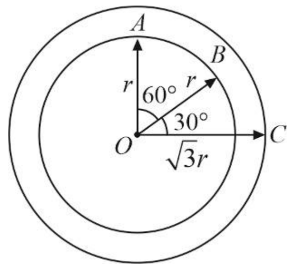
Candidate's Signature : _ _ _ _ _ _ _ _ _ _ **Invigilator's Signature** _ _ _ _ _ _ _ _

PART I : PHYSICS 100 MARKS

SECTION-1

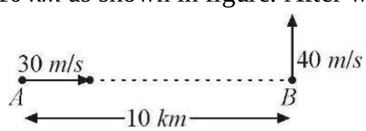
This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. The resultant of the three vectors $\overline{OA}, \overline{OB}$ and $\overline{OC}$ shown in adjoining figure is:



(A) r **(B)** $3r$ **(C)** $r(1+\sqrt{2})$ **(D)** $r(\sqrt{2}-1)$

2. Ship A is moving with velocity 30 m/s due east and ship B with velocity 40 m/s due north. Initial separation between the ships is 10 km as shown in figure. After what time ships are closest to each other?

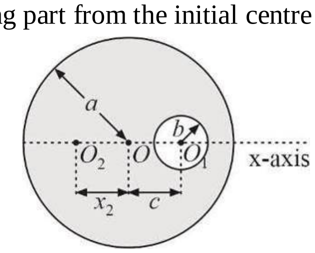


(A) 80 s **(B)** 120 s **(C)** 160 s **(D)** 100 s

3. The velocity of a small ball of mass M and density d_1 becomes constant finally, when dropped in a container filled with glycerine. If the density of glycerine is d_2, the viscous force acting on ball after long time is:

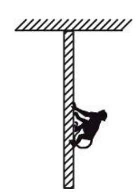
(A) $\dfrac{Md_1g}{d_2}$ **(B)** $Mg\left(1-\dfrac{d_2}{d_1}\right)$ **(C)** $\dfrac{M(d_1+d_2)}{g}$ **(D)** $Mg\left(1-\dfrac{d_1}{d_2}\right)$

4. A uniform circular disc of radius a is taken. A circular portion of radius b has been removed from it as shown in the figure. If the centre of hole is at a distance c from the centre of the disc, the distance x_2 of the centre of mass of the remaining part from the initial centre of mass O is given by:



(A) $\dfrac{\pi b^2}{(a^2-b^2)}$ **(B)** $\dfrac{-cb^2}{(a^2-b^2)}$ **(C)** $\dfrac{\pi c^2}{(a^2-b^2)}$ **(D)** $\dfrac{\pi a^2}{(c^2-b^2)}$

5. A uniform rope of mass m hangs freely from a ceiling. A monkey of mass M climbs up the rope with an acceleration a. The force exerted by the rope on the ceiling is:



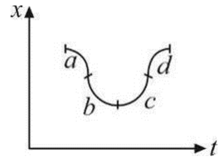
(A) $Ma + mg$

(B) $M(a + g) + mg$

(C) $M(a + g)$

(D) Dependent on the position of monkey on the rope

6. The graph shown is a plot of position versus time. For which labeled region is the velocity positive and the acceleration negative? (Velocity/acceleration is taken as positive if it is directed along +x axis)



(A) a **(B)** b **(C)** c **(D)** d

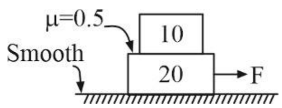
7. A disc of mass m and radius R has a concentric hole of radius r. Its moment of inertia about an axis through its centre and perpendicular to its plane is:

(A) $\frac{1}{2}m(R-r)^2$ **(B)** $\frac{1}{2}m(R^2-r^2)$ **(C)** $\frac{1}{2}m(R+r)^2$ **(D)** $\frac{1}{2}m(R^2+r^2)$

8. For a particle moving along +x-axis, acceleration is given as $a = 2v^2$, where v is its instantaneous speed. If the speed of the particle is v_0 at $x = 0$, find speed as a function of x.

(A) $v = v_0 e^{2x}$ **(B)** $v = v_0 e^{x}$ **(C)** $v = v_0 \ln 2x$ **(D)** $v = v_0 \ln x$

9. Initially, the system is at rest. Find out maximum value of F for which the blocks move together. [Take $g = 10\ m/s^2$]



(A) 200 N **(B)** 150 N **(C)** 100 N **(D)** 50 N

10. A body of mass m when released from rest from a height h, hits the ground with speed $\sqrt{gh}$. Find work done by air resistance force.

(A) $-mgh$ **(B)** mgh **(C)** $-\frac{mgh}{2}$ **(D)** $\frac{mgh}{2}$

11. A body is projected vertically upwards with speed $\sqrt{gR_e}$ from the surface of earth. What is the maximum height attained by the body? [R_e is the radius of earth, g is acceleration due to gravity]

(A) R_e **(B)** $2R_e$ **(C)** $\frac{R_e}{2}$ **(D)** None of these

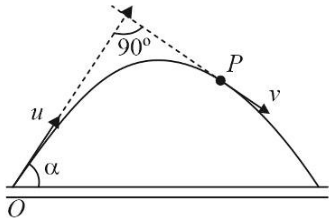
12. A spring balance extends by 2 cm due to a mass on surface of Earth. What will be the extension at a height $2R_e$ from earth's surface?

(A) $\frac{1}{2}cm$ **(B)** $\frac{1}{4}cm$ **(C)** $\frac{2}{9}cm$ **(D)** $\frac{1}{9}cm$

13. A rocket starts from rest and moves up with acceleration 2g due to thrust force by the ejected gases. After time t_0, its engine is switched off after which it goes into free fall. What will be the maximum height attained by rocket, from ground? (Neglect variation of g with height)

(A) $4gt_0^2$ **(B)** $3gt_0^2$ **(C)** $2gt_0^2$ **(D)** gt_0^2

14. A particle is projected from point *O* with velocity *u* in a direction making an angle α with the horizontal. At an instant it is at point *P*, where its velocity is at right angles to the initial direction of projection. Its velocity at point P is:

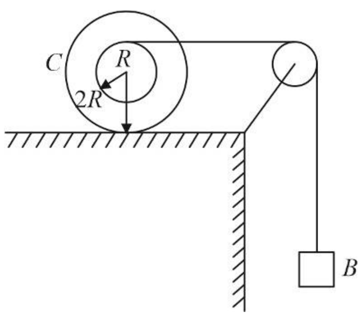


(A) $u\tan\alpha$ **(B)** $u\cot\alpha$ **(C)** $u\,\text{cosec}\,\alpha$ **(D)** $u\sec\alpha$

15. A particle moves in a circle of radius 2 cm at a speed given by $v = 4t$, where v is in cms^{-1} and *t* is in seconds. Find the acceleration of particle at $t = 1s$.

(A) $4\ cm/s^2$ **(B)** $8\ cm/s^2$ **(C)** $4\sqrt{5}\ cm/s^2$ **(D)** None of these

16. In the figure, mass of stepped cylinder *C* and block *B* is *m* each. Moment of inertia of *C* about its central axis is $2mR^2$. C rolls on the horizontal surface while B moves down. There is no slipping between any surfaces in contact. The ratio of kinetic energy of *C* to that of block *B* is:

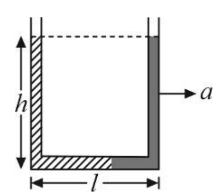


(A) $\frac{3}{4}$ **(B)** $\frac{1}{3}$ **(C)** $\frac{2}{3}$ **(D)** $\frac{1}{2}$

17. A cylindrical tank has a hole of area 1 cm^2 in its bottom. If water is allowed to flow into the tank from a tube above it at the rate of 70 cm^3/s, then the maximum height up to which water can rise in the tank is:

(A) 2.5 cm **(B)** 5 cm **(C)** 10 cm **(D)** 0.25 cm

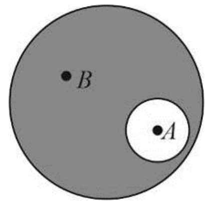
18. A *U*-tube of base length '*l*' filled with same volume of two liquids of densities ρ and 2ρ is moving with an acceleration '*a*' on the horizontal plane. If the height difference between the two surfaces (open to atmosphere) becomes zero, then the height *h* is given by:



(1) $\frac{a}{2g}l$ **(2)** $\frac{3a}{2g}l$

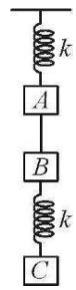
(3) $\frac{a}{g}l$ **(4)** $\frac{2a}{3g}l$

19. There is an air bubble of radius R inside a drop of water of radius $3R$. Find the ratio of gauge pressure at point B to that at point A.



(A) $\frac{1}{2}$ **(B)** $\frac{1}{4}$ **(C)** $\frac{1}{3}$ **(D)** 1

20. The system shown in the given figure is in equilibrium. All the blocks are of the equal mass m and springs are ideal. When the string between A and B is cut then immediately after that:



(1) The acceleration of block A is 2 g while the acceleration of block C is g
(2) The acceleration of block B is 2 g while the acceleration of block C is zero
(3) The acceleration of block A is 2 g while the acceleration of block B is g
(4) The acceleration of block B is g while the acceleration of block C is zero

SPACE FOR ROUGH WORK

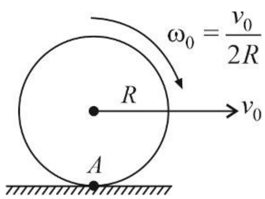
SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. ***In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*****. (Example: 6, 81, 1.50, 3.25, 0.08)**

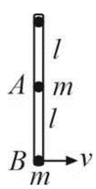
21. A big particle of mass $(3 + m)$ kg is initially at rest. It blasts into 3 pieces, such that a particle of mass 1 kg moves along x-axis with velocity 2 m/s and a particle of mass 2 kg moves with velocity 1 m/s perpendicular to direction of 1 kg particle. If the third particle moves with velocity $\sqrt{2}\ m/s,$ then m (in Kg) is_____ .

22. A particle of mass 1 kg is moving with a uniform speed of 4 m/sec in $x - y$ plane along the line $3y = 4x + 10$. The magnitude of its angular momentum about the origin in $kg - m^2/s$ is _______.

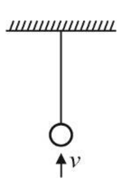
23. A hollow sphere of mass M and radius R as shown in figure slips on a rough horizontal plane. At some instant it has linear velocity v_0 and angular velocity about the centre $\frac{v_0}{2R}$ as shown in figure. If the linear velocity after the sphere starts pure rolling is $\frac{4v_0}{n}$, then n is______.



24. Two-point masses of mass m are connected to a light rod of length l hinged at its top end. It is free to rotate in vertical plane, as shown. If the minimum horizontal velocity v is given to mass B so that rod completes the circular motion in vertical plane is $\sqrt{xgl}$, then x is______.



25. A bullet of mass 50 g is fired from below into a bob of mass 450 g of a long simple pendulum as shown. The bullet remains inside the bob and the bob rises through a height of 1.8 m. The speed v of the bullet was______(in m/s).



SPACE FOR ROUGH WORK

PART II : CHEMISTRY 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. 10.1g of KNO_3 is dissolved in 500 mL of H_2O. Mass of $Ba(NO_3)_2$ that should be added to this solution to get a molality (m) of 0.3 with respect to NO_3^- ion is :

(Mw of $KNO_3 = 101\text{ g mol}^{-1}$, Mw of $Ba(NO_3)_2 = 261\text{ g mol}^{-1}, dH_2O = 1\text{ gml}^{-1}$)

(A) $\approx 1.3g$ **(B)** $\approx 13g$ **(C)** $\approx 6.5g$ **(D)** $\approx 65g$

2. Arrange the following in order of increasing masses.

(i) 1 molecule of oxygen **(ii)** 1 atom of nitrogen

(iii) 1 mol of water **(iv)** 1×10^{-10} g of iron

[Atomic Mass : O = 16, N = 14, H = 1, Fe = 56]

(A) ii < i < iii < iv **(B)** i < ii < iv < iii **(C)** ii < i < iv < iii **(D)** i < ii < iii < iv

3. An electron is continuously accelerated in a vacuum tube by applying potential difference. If its de-Broglie wavelength is decreased by 1%, the change in the kinetic energy of the electron is nearly.

(A) Decreased by 1.0% **(B)** Increased by 2.0%

(C) Increased by 1.0% **(D)** Decreased by 2.0%

4. Ratio of frequency of revolution of electron in the second excited state of He^+ and second state of hydrogen is :

(A) $\frac{32}{27}$ **(B)** $\frac{27}{32}$ **(C)** $\frac{1}{54}$ **(D)** $\frac{27}{2}$

5. Which of the following statements about the electromagnetic spectrum is not correct ?

(A) IR radiations have larger wavelength than cosmic rays

(B) The frequency of microwaves is less than that of UV rays

(C) X-rays have larger wavenumber than microwaves

(D) The velocity of X-rays is more than that of microwaves

6. Which of the following gases will have the highest RMS velocity at 25°C ?

[Atomic Mass : O = 16, C = 12, S = 32]

(A) O_2 **(B)** CO_2 **(C)** SO_2 **(D)** CO

7. If for two gases of molecular weights M_A and M_B at temperatures T_A and T_B, $T_AM_B = T_BM_A$, then which of the following properties has the same magnitude for both the gases ?

(A) Density **(B)** Pressure **(C)** KE per mole **(D)** u_{rms}

8. An ideal gas obeying the kinetic theory of gases can be liquefied if :

(A) Its temperature is more than its critical temperature (T_c)

(B) Its pressure is more than its critical pressure (P_c)

(C) Its pressure is more than P_c at a temperature less than T_c

(D) It cannot be liquefied at any value of P and T

9. What is the value of b (van der Waals constant) if the diameter of a molecule is $2.0\,\text{Å}$?

(Use $\pi = 3.14$, $N_A = 6 \times 10^{23}$)

(A) ≈ 2.4 mL mol^{-1} **(B)** ≈ 4.8 mL mol^{-1}

(C) ≈ 7.2 mL mol^{-1} **(D)** ≈ 10 mL mol^{-1}

10. During reversible adiabatic expansion, a gas obeys $VT^3 = \text{constant}$. The gas must be :

(A) Monoatomic **(B)** Diatomic **(C)** Polyatomic **(D)** Data insufficient

11. At 298 K, ΔH_f° of methanol is given by the chemical equation:

(A) $CH_4(g) + \frac{1}{2}O_2(g) \longrightarrow CH_3OH(g)$

(B) $C(\text{graphite}) + \frac{1}{2}O_2(g) + 2H_2(g) \longrightarrow CH_3OH(\ell)$

(C) $C(\text{diamond}) + \frac{1}{2}O_2(g) + 2H_2(g) \longrightarrow CH_3OH(\ell)$

(D) $CO(g) + 2H_2(g) \longrightarrow CH_3OH(\ell)$

12. Considering entropy (S) as a thermodynamic parameter, the criterion for the spontaneity of any process is :

(A) $\Delta S_{sys} + \Delta S_{surr} > 0$ **(B)** $\Delta S_{sys} - \Delta S_{surr} > 0$

(C) $\Delta S_{sys} > 0$ only **(D)** $\Delta S_{surr} > 0$ only

13. The heat of combustion of solid benzoic acid at constant volume is $-321.30\,\text{kJ}$ at 27°C. The heat of combustion at constant pressure is :

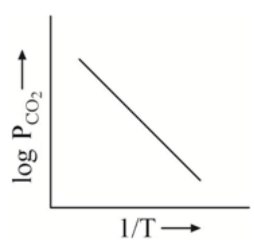
(A) $-321.30 - 300\,R$ **(B)** $-321.30 + 300\,R$

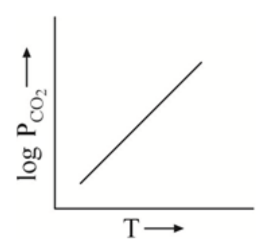
(C) $-321.30 - 150\,R$ **(D)** $-321.30 - 900\,R$

14. For the chemical equilibrium,

$$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$$

$\Delta_r H^\ominus$ can be determined from which one of the following plots ?

(A)


(B)


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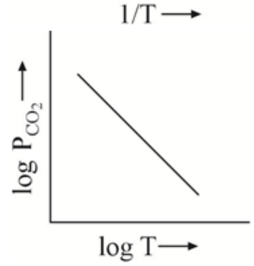
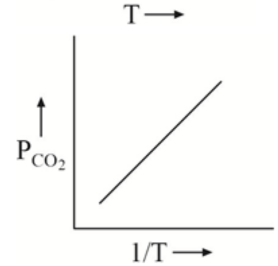


(D)


15. For the reaction $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$.

The degree of dissociation (α) of $HI(g)$ is related to equilibrium constant K_p by the expression.

(A) $\dfrac{1+2\sqrt{K_p}}{2}$ **(B)** $\sqrt{\dfrac{1+2K_p}{2}}$ **(C)** $\sqrt{\dfrac{2K_p}{1+2K_p}}$ **(D)** $\dfrac{2\sqrt{K_p}}{1+2\sqrt{K_p}}$

16. $Au(s) \rightleftharpoons Au(\ell)$ above mentioned equilibrium is favoured at :

(A) High pressure, low temperature **(B)** High pressure, high temperature
(C) Low pressure, high temperature **(D)** Low pressure, low temperature

17. Compare the percentage s-characthter in the hybrid orbital of B and N in BF_3 and NF_3 respectively:

(A) % s character is same in both B and N **(B)** % s character in B is more than N
(C) % s character in N is more than B **(D)** Can't be predicted

18. Given the species N_2, CO, NO^+ and CN^-, which of the following statements are true for this :

(I) All the species are diamagnetic
(II) All the species are isostructural
(III) All the species have identical bond order
(IV) More than one species have zero dipole order

(A) I, II and III **(B)** I, II, III and IV **(C)** III and IV **(D)** I and II

19. The IUPAC name of tertiary butyl iodide is :

(A) 1-iodo-3-methyl propane **(B)** 2-iodo-2-methyl propane
(C) 4-iodobutane **(D)** 2-iodobutane

20. The value of enthalpy change (ΔH) for the reaction $C_2H_5OH(\ell) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(\ell)$ is -1366.5 kJ mol^{-1}. The value of internal energy change for the above reaction at this temperature will be:

(A) –1371.5 kJ **(B)** –1369.0 kJ **(C)** –1364.0 kJ **(D)** –1361.5 kJ

SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)

21. Calculate the mass of carbon present in 0.1 mole of sodium ferricyanide $Na_3[Fe(CN)_6]$.
[Atomic Mass : Na = 23, C = 12, N = 14, Fe = 56]

22. Supposing that the Pauli exclusion principle is not correct, if one orbital can accommodate three electrons. According to new system, number of electrons having $n = 2$ for second member of noble gas family are____.

23. At 627°C and 1 atm SO_3 is partially dissociated into SO_2 and O_2 by the reaction

$$SO_3(g) \rightleftharpoons SO_2(g) + \frac{1}{2}O_2(g)$$

The density of the equilibrium mixture is $0.925\ g\ L^{-1}$. What is the degree of dissociation?

(Use $R = 0.0821\ L\ atm\ K^{-1}\ mol^{-1}$, Atomic Mass : S = 32, O = 16)

24. How many unpaired electron(s) is/are present in O_2 molecule if Hund's rule is violated ?

25. Find the number of molecule(s) is/are having zero dipole moment and linear shape from the following :
ClF_3, CO_2, $HgCl_2$, ICl_2^-, XeF_2, BeF_2

SPACE FOR ROUGH WORK

PART III : MATHEMATICS 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Locus of mid-point of chords of $x^2+y^2+2gx+2fy+c=0$ that pass through the origin, is :

(A) $x^2+y^2+2gx+2fy=0$ **(B)** $x^2+y^2+gx+fy+c=0$

(C) $x^2+y^2+gx+fy=0$ **(D)** $2\left(x^2+y^2+gx+fy\right)+c=0$

2. Tangents PA and PB drawn to $x^2+y^2=9$ from any arbitrary point 'P' on the line $x+y=25$. Locus of mid-point of chord AB is :

(A) $25\left(x^2+y^2\right)=9(x+y)$ **(B)** $25\left(x^2+y^2\right)=3(x+y)$

(C) $5\left(x^2+y^2\right)=3(x+y)$ **(D)** None of these

3. The value of $S=\sum_{k=1}^{6}\left(\sin\frac{2\pi k}{7}-i\cos\frac{2\pi k}{7}\right)$ is :

(A) -1 **(B)** 0 **(C)** $-i$ **(D)** i

4. If $|z|=1, z\neq 1$, then value of principal $arg\left(\frac{1}{1-z}\right)$ cannot exceed:

(A) $\frac{\pi}{2}$ **(B)** 0 **(C)** $\frac{\pi}{3}$ **(D)** $\frac{\pi}{4}$

5. If α,β are the roots of $x^2+px+q=0$, and ω is a cube root of unity, then value of $\left(\omega\alpha+\omega^2\beta\right)\left(\omega^2\alpha+\omega\beta\right)$ is :

(A) p^2 **(B)** $3q$ **(C)** p^2-2q **(D)** p^2-3q

6. If z_1,z_2,z_3 are complex numbers such that $|z_1|=|z_2|=|z_3|=\left|\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}\right|=1$. Then $|z_1+z_2+z_3|$ is :

(A) Equal to 1 **(B)** Less than 1 **(C)** Greater than 3 **(D)** Equal to 3

7. Tangents are drawn to $x^2+y^2=1$ from any arbitrary points P on the line $2x+y-4=0$. The corresponding chord of contact passes through a fixed point whose coordinates are

(A) $\left(\frac{1}{4},\frac{1}{2}\right)$ **(B)** $\left(\frac{1}{2},1\right)$ **(C)** $\left(\frac{1}{2},\frac{1}{4}\right)$ **(D)** $\left(1,\frac{1}{2}\right)$

8. The value of $\frac{(\cos 2\theta-i\sin 2\theta)^7(\cos 3\theta+i\sin 3\theta)^{-5}}{(\cos 4\theta+i\sin 4\theta)^{12}(\cos 5\theta+i\sin 5\theta)^{-6}}$ is :

(A) $\cos 33\theta+i\sin 33\theta$ **(B)** $\cos 33\theta-i\sin 33\theta$

(C) $\cos 47\theta+i\sin 47\theta$ **(D)** $\cos 47\theta-i\sin 47\theta$

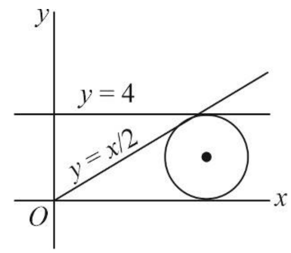
9. If z is a complex number of unit modulus and argument θ, then $arg\left(\frac{1+z}{1+\bar{z}}\right)$ is equal to :

(A) $-\theta$ **(B)** $\frac{\pi}{2}-\theta$ **(C)** θ **(D)** $\pi-\theta$

10. If $x+y=k$ is a normal to the parabola $y^2=12x$, p is the length of the perpendicular from the focus of the parabola on this normal, then $3k^3+2p^2$ is equal to :

(A) 2223 **(B)** 2224 **(C)** 2222 **(D)** None of these

11. The x-coordinate of the center of the circle in the first quadrant (see figure) tangent to the lines $y=\frac{1}{2}x,\ y=4$ and the x-axis is:



(A) $4+2\sqrt{5}$ **(B)** $4+\frac{8\sqrt{5}}{5}$

(C) $2+\frac{6\sqrt{5}}{5}$ **(D)** $8+2\sqrt{5}$

12. The axis of a parabola is along the line $y = x$ and the distance of its vertex from the origin is $\sqrt{2}$ and that of its focus from the origin is $2\sqrt{2}$. If the vertex and focus lie in the first quadrant, then equation of the parabola is :

(A) $(x+y)^2 = x-y-2$ **(B)** $(x-y)^2 = x+y-2$

(C) $(x-y)^2 = 4(x+y-2)$ **(D)** $(x-y)^2 = 8(x+y-2)$

13. The equation of normal to the ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$ at the end of the latus rectum in first quadrant is: (e is eccentricity of ellipse)

(A) $x-ey-2e^3=0$ **(B)** $x+ey+2e^3=0$

(C) $x+2ey+e^3=0$ **(D)** $x-2ey+e^3=0$

14. The parameters t and t' of two points on the parabola $y^2=4ax$, are connected by the relation $t=k^2t'$. The tangents at their points intersect on the curve :

(A) $y^2=ax$ **(B)** $y^2=k^2x$ **(C)** $y^2=ax\left(k+\frac{1}{k}\right)^2$ **(D)** None of these

15. If the tangent at the point ϕ on the ellipse $\frac{x^2}{16}+\frac{11y^2}{256}=1$ touches the circle $x^2+y^2-2x-15=0$, then ϕ is equal to :

(A) $\pm\frac{\pi}{2}$ **(B)** $\pm\frac{\pi}{4}$ **(C)** $\pm\frac{\pi}{3}$ **(D)** $\pm\frac{\pi}{6}$

16. The length of the intercept on y-axis cut off by the parabola, $y^2-5y=3x-6$ is:
(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

17. Tangents are drawn to the circle $x^2+y^2=1$ at the points where it is met by the circles, $x^2+y^2-(\lambda+6)x+(8-2\lambda)y-3=0,\ \lambda$ being the variable. The locus of the point of intersection of these tangents is:
(A) $2x-y+10=0$ **(B)** $x+2y-10=0$ **(C)** $x-2y+10=0$ **(D)** $2x+y-10=0$

18. Equation of the smallest circle passing through A and B, where A and B are intersection points of circles $x^2+y^2-4x-4y+4=0$ and $x^2+y^2-x-y+\frac{1}{4}=0$ is :

(A) $\left(x-\frac{3}{8}\right)^2+\left(y-\frac{3}{8}\right)^2=\frac{7}{32}$ **(B)** $(x-1)^2+(y-1)^2=7^2$

(C) $(x-2)^2+(y-2)^2=1$ **(D)** $\left(x-\frac{5}{8}\right)^2+\left(y-\frac{5}{8}\right)^2=\frac{7}{32}$

19. If $z=-2+2\sqrt{3}i$, then $z^{2n}+2^{2n}z^n+2^{4n}$ (where n is a positive integer) cannot be equal to :
(A) 54 **(B)** 0
(C) $3\cdot 4^{2n}$, where n is multiple of 3 **(D)** none of these

20. If $z_r = cos\frac{r\alpha}{n^2} + i\,sin\frac{r\alpha}{n^2}$, where $r = 1, 2, 3, ..., n$, then $\prod_{r=1}^{\infty}(z_r)$ is equal to :

(A) $cos\alpha + i\,sin\alpha$ **(B)** $cos\left(\frac{\alpha}{2}\right) - i\,sin\left(\frac{\alpha}{2}\right)$ **(C)** $e^{i\alpha/2}$ **(D)** $\sqrt[3]{e^{i\alpha}}$

SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)

21. Let $\omega = e^{2\pi i/3}$, and a, b, c, x, y, z be non-zero complex numbers such that :

$$a + b + c = x,$$
$$a + b\omega + c\omega^2 = y$$
$$a + b\omega^2 + c\omega = z$$

Then the value of $\frac{|x|^2 + |y|^2 + |z|^2}{|a|^2 + |b|^2 + |c|^2}$ is equal to _________.

22. If $z^2 + z + 1 = 0$, where z is a complex number, then value of

$S = \left(z + \frac{1}{z}\right)^2 + \left(z^2 + \frac{1}{z^2}\right)^2 + \left(z^3 + \frac{1}{z^3}\right)^2 + ... + \left(z^6 + \frac{1}{z^6}\right)^2$ is _________.

23. If $x + iy = \frac{3}{cos\theta + i\,sin\theta + 2}$, then $4x - x^2 - y^2$ reduces to _________.

24. The number of common tangents to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x - 8y = 24$ is _________.

25. The length of the common tangent to the ellipse $\frac{x^2}{25} + \frac{y^2}{4} = 1$ and the circle $x^2 + y^2 = 16$ intercept by the coordinate axes is _________.

SPACE FOR ROUGH WORK

End of JEE Main - 6 | JEE 2022